

LanPro — Dokumen Gabungan BRD · FSD · TSD

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CONTENTS

LanPro — Dokumen Gabungan BRD · FSD · TSD	
Daftar isi	
1. BRD — Business Requirements	
1.1 Masalah bisnis	
1.2 Solusi produk (posisi resmi §24.2 / #346)	
1.3 Tujuan & KPI	
1.4 Pemangku kepentingan & persona	
Tier A — Peran sistem (Users.role)	
Tier B — Peran proyek (ProjectMembers.role)	
1.5 Ruang lingkup	
1.6 Asumsi, risiko, ketergantungan	
1.7 Aturan bisnis lintas-produk (katalog BR-*)	
2. FSD — Functional Specification	
2.1 Modul fungsional (sidebar)	
2.2 Persyaratan non-fungsional	
3. TSD — Technical Specification	
3.1 Arsitektur runtime	
3.2 Lapisan penting	
3.3 Alur request otorisasi	
3.4 Stack & gerbang	
3.5 Integrasi	
4. Flowchart — Alur sistem LanPro	
4.1 Peta modul end-to-end	

4.2 Siklus hidup akun	
4.3 Siklus hidup isu	
5. Flowchart — Agile vs Waterfall	
5.1 Pemilihan metodologi	
5.2 Alur Agile	
5.3 Alur Waterfall	
5.4 Perbandingan cepat	
6. Flowchart — Per peran sistem	
6.1 Administrator (admin)	
6.2 Department Head (head sistem)	
6.3 Standard User & Observer sistem	
7. Flowchart — Per peran proyek	
7.1 Owner / Project Admin / Manager	
7.2 System Analyst / Business Analyst	
7.3 Developer / QA / Head / Viewer	
8. Matriks akses	
8.1 Sistem	
8.2 Proyek (cuplikan)	
9. Spesifikasi per modul	
9.1 Auth & akun	
9.2 Proyek (buat, setelan, metodologi)	
9.3 Dashboard	
9.4 Daftar Isu / Task	
9.5 Planning & Sprint (Agile)	
9.6 Papan Kanban	

9.7 Timeline / Roadmap / Milestone	
9.8 QA	
9.9 Wiki (Dokumentasi)	
9.10 Flowchart	
9.11 Tim / Access	
9.12 Administrasi sistem	
9.13 Catatan Rapat (modul induk Meeting AI)	
10. Meeting AI — spesifikasi dalam	
10.1 BRD Meeting AI	
10.2 FSD — tiga jalur input	
10.3 TSD — komponen & API	
API utama	
Env & dependensi	
10.4 State machine pipeline	
10.5 Swimlane end-to-end (live/upload)	
10.6 Swimlane manuscript (paste)	
10.7 Progress & kegagalan	
10.8 Gap & risiko (#320)	
11. Auth & OIDC — spesifikasi dalam	
11.1 BRD Auth	
11.2 FSD Auth	
11.3 TSD Auth	
OIDC flow	
Password forgot (ringkas terkini)	
11.4 Gap Auth	

12. Lampiran

12.1 Artefak terkait

12.2 Yang dokumen ini bukan

12.3 Riwayat versi

12.4 Keputusan review tersisa

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LanPro — Dokumen Gabungan BRD · FSD · TSD

Meta	Nilai
Produk	LanPro — manajemen proyek + kolaborasi (Issue, Sprint/Milestone, Kanban, QA, Wiki, Flowchart, Meeting AI)
Versi dokumen	3.1.0 (06 Sep 2026)
Status	DRAFT review pemilik — selaras kode + <code>AUDIT.md</code> §19 + papan 7 BELUM · 454 SELESAI (+ item #466)
Tiket papan	#466 — dokumentasi BRD/FSD/TSD lengkap (per modul, Meeting AI, OIDC, flowchart)
Keputusan pemilik	(1) setuju #466 · (2) flowchart per modul · (3) PDF · (4) alur Meeting AI + detail bisnis/teknis
Sumber RBAC	<code>src/lib/matriksAkses.ts</code> ↔ <code>AUDIT.md</code> §19.4 / §19.5
Metodologi	Dual-mode Agile / Waterfall (#311, #312, #465)
Bahasa	Indonesia (istilah teknis EN dipertahankan)
v2.5 diganti (RBAC basi). v3.0 = kerangka. v3.1 = detail per modul + Meeting AI + Auth/OIDC + swimlane.	

Daftar isi

1. [BRD — Business Requirements](#)
2. [FSD — Functional Specification \(ringkas lintas modul\)](#)
3. [TSD — Technical Specification](#)
4. [Flowchart — Alur sistem LanPro](#)
5. [Flowchart — Agile vs Waterfall](#)
6. [Flowchart — Per peran sistem](#)
7. [Flowchart — Per peran proyek](#)
8. [Matriks akses](#)
9. [Spesifikasi per modul \(BRD+FSD+TSD+flow\)](#)

10. [Meeting AI — spesifikasi dalam](#)
 11. [Auth & OIDC — spesifikasi dalam](#)
 12. [Lampiran](#)
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1. BRD — Business Requirements

1.1 Masalah bisnis

Organisasi engineering sering memakai **4–6 tool terpisah** (tracker, board, dokumen, QA spreadsheet, notulen, diagram). Akibatnya:

1. Silo requirement ↔ development ↔ QA ↔ notulen.
2. Hak akses tidak seragam (risiko IDOR / over-permission).
3. Master data (status, peran, departemen) tidak kanonik — label vs code bercampur.
4. Metodologi **Agile** dan **Waterfall** dicampur tanpa aturan navigasi yang jernih.
5. Notulen rapat tidak otomatis menjadi poin diskusi / kandidat isu.

1.2 Solusi produk (posisi resmi §24.2 / #346)

LanPro = **pelacak agile-first** (backlog, sprint, kanban, poin, burndown) **plus** Gantt/milestone, wiki, rapat AI, QA, flowchart — **bukan** hibrida Jira + MS Project penuh.

Metodologi	Perilaku produk
Agile	Menu Planning & Sprint; scope lock; satu sprint aktif; burndown hybrid poin tugas
Waterfall	Menu Sprint disembunyikan ; Roadmap & Milestones; create sprint API ditolak

Ditolak secara produk (#346): CPM/critical path, WBS penuh, timesheet kapasitas, marketplace automation Jira-class.

1.3 Tujuan & KPI

Tujuan	KPI	Status / catatan
Satu workspace SDLC	Modul inti dipakai 1 org	Issues + Sprint/Roadmap + Meeting + Wiki + QA
Keamanan deny-by-default	0 rute proyek tanpa <code>jagaProyek</code>	Two-Tier RBAC F7
Dual-mode bersih	Copy Sprint ≠ Phase; nav berubah	#465 SELESAI kode
Meeting → insight	Pipeline sampai COMPLETED	#320 BELUM E2E pemilik
Durabilitas unggahan	Storage bukan ephemeral di production	#30 terbuka

1.4 Pemangku kepentingan & persona

TIER A — PERAN SISTEM (`USERS.ROLE`)

Code	Label	Tanggung jawab bisnis
<code>admin</code>	Administrator	Onboarding user, master data, audit, buat proyek, God Mode lintas proyek
<code>head</code>	Department Head	Baca admin terbatas; daftar proyek se-departemen
<code>user</code>	Standard User	Anggota proyek sesuai <code>ProjectMembers</code>
<code>viewer</code>	Observer (sistem)	Baca terbatas di lingkup sistem

TIER B — PERAN PROYEK (`PROJECTMEMBERS.ROLE`)

Code	Label	Wilayah kuasa bisnis
<code>owner</code>	Project Owner	Pemilik; hapus proyek; setelan
<code>admin</code>	Project Admin	Administrasi proyek (bukan God Mode sistem)
<code>manager</code>	Project Manager	Perencanaan, isu, board, sprint/timeline
<code>head</code>	Department Head (proyek)	Baca seluruh modul proyek
<code>system_analyst</code>	System Analyst	CRUD Wiki + Flowchart
<code>business_analyst</code>	Business Analyst	CRUD Meeting Notes + AI
<code>developer</code>	Developer	CRU isu; RU board/QA
<code>qa</code>	QA	CRUD modul QA; taut bug → Task
<code>viewer</code>	Viewer (proyek)	Baca saja

1.5 Ruang lingkup

In scope: auth (email/OIDC), proyek, isu/task, sprint/milestone, kanban, roadmap/Gantt, QA, wiki, flowchart, meeting (+ AI pipeline), tim/akses, master data, audit, dual-mode, RBAC dua tingkat.

Out of scope: CPM, WBS penuh, timesheet, marketplace automation, parity Otter.ai penuh (diarization speaker masih gap #320).

1.6 Asumsi, risiko, ketergantungan

Item	Risiko / asumsi	Tiket
Storage local di Vercel	File ephemeral	#30 · #290
Meeting recording lokal	Tidak memakai <code>STORAGE_DRIVER</code> S3	#30 · #320
Bukti visual tab bersih	Gerbang ≠ bukti UI	#335
Meeting AI COMPLETED	Butuh Gemini + ffmpeg + E2E pemilik	#320
SSO Microsoft lapangan	Adaptor ada; verifikasi domain	#305
Soft-delete	Ditolak — hanya hard-delete + konfirmasi	#394

1.7 Aturan bisnis lintas-produk (katalog BR-*)

ID	Aturan	Referensi
BR-RBAC-01	Deny-by-default; God Mode hanya <code>Users.role=admin</code> + audit	§19.6
BR-METH-01	Waterfall: POST sprint ditolak; menu Sprint disembunyikan	#311 #465
BR-SPR-01	Lingkup sprint aktif/selesai terkunci kecuali <code>unlockScope</code>	#461
BR-SPR-02	Maksimal satu sprint <code>active</code> per proyek	#462
BR-MET-01	Burndown: poin bila <code>totalPoints>0</code> else hitungan tugas	#464
BR-QA-01	Bug QA: <code>linkedTaskId</code> + <code>linkedBugKey</code> tampilan	#463
BR-DEL-01	Soft-delete ditolak; hard-delete + konfirmasi + cascade	#394 #444
BR-IRIS-01	Tidak membangun CPM/timesheet marketplace	#346
BR-MD-01	Yang disimpan ke DB adalah code MasterData, bukan label	AGENTS.md
BR-MEET-01	Rekaman hanya untuk analisis; hapus file setelah COMPLETED	#320
BR-MEET-02	PUT/DELETE meeting: author atau system ADMIN (lebih ketat dari matriks)	meetings.routes

2. FSD — Functional Specification

2.1 Modul fungsional (sidebar)

Modul UI	Key matriks	Fungsi utama
Dashboard	dashboard	KPI, burndown/velocity (Agile), ringkasan
Daftar Isu	list	CRUD isu, History, lampiran, komentar
Perencanaan & Sprint	sprints	Backlog ↔ sprint, start/complete, scope lock
Papan Kanban	board	DnD status MasterData, WIP lunak
Peta Jalan & Linimasa	timeline	Gantt, milestone, garis blocks
Penilaian Kualitas	qa	Suite/case, eksekusi, bug → Task
Dokumentasi	wiki	Dokumen proyek
Editor Diagram Alur	flowchart	Kanvas flowchart
Catatan Rapat	meetingNotes	Notulen + Meeting AI
Tim	access	Anggota & peran proyek
Master / Users / Audit / DB / Settings	modul sistem	Administrasi

Detail per modul: §9. Meeting AI: §10. Auth: §11.

2.2 Persyaratan non-fungsional

- UI token `surface-*` / `content-*` / `border-*` (§22); gerbang `audit:tema` / `audit:warna`.
- i18n ID/EN.
- PostgreSQL saja (Neon pooler); identifier camelCase dikutip.
- JWT di klien (arah httpOnly terkait #30).
- Hard-delete saja (#394).

3. TSD — Technical Specification

3.1 Arsitektur runtime

```

Browser (React 19 + Vite + Tailwind v4)
  | HTTPS + Socket.IO
Express (server.ts) :3000
  | jagaProyek / verifyGlobalAdmin / JWT
Routes → Controllers/Services → Repositories → PostgreSQL (Neon)
  | optional Redis (Socket.IO adapter)
  | Gemini (Meeting AI) · FFmpeg · local uploads · Resend/SMTP

```

3.2 Lapisan penting

Lapisan	Path	Catatan
Fitur UI	src/features/*	Satu folder per domain
Primif UI	src/components/	Stateless lintas fitur
Matriks RBAC	src/lib/matriksAkses.ts	Sinkron AUDIT §19
Peran	src/types/roles.ts	Dua enum SYSTEM/PROJECT
Metodologi	src/lib/methodology.ts , alurMetodologi.ts	
Sprint lingkup	server/lib/sprintLingkup.ts	#461
Migrasi	src/lib/pg-migrate.ts	Jangan sentuh src/lib/db.ts sembarangan
Penjaga proyek	server/middleware/jagaProyek.ts	

3.3 Alur request otorisasi

```

flowchart TD
    A[HTTP /api/...] --> B{Rute publik?}
    B -->|Ya| C[Handler]
    B -->|Tidak| D[authenticateJWT]
    D --> E{God Mode admin sistem?}
    E -->|Ya + audit| C
    E -->|Tidak| F{Anggota proyek?}
    F -->|Tidak| G[403]
    F -->|Ya| H[jagaProyek modul,aksi]
    H --> I{bolehDiProyek?}
    I -->|Tidak| G
    I -->|Ya| C
  
```

3.4 Stack & gerbang

- Dev: `npm run dev` → `tsx watch server.ts`
- Build: Vite FE + esbuild `dist/server.cjs`
- Gerbang: `doctor`, `lint`, `test`, `build`, `audit:papan|warna|tema`
- UI benar = **tab peramban bersih**, bukan hanya build hijau (§15.3)

3.5 Integrasi

Integrasi	Fungsi	Env kunci
Neon PostgreSQL	Data primer	<code>DATABASE_URL</code> (pooler)
Google Gemini	Meeting AI STT + LLM	<code>GEMINI_API_KEY</code>
OIDC Google / Microsoft	SSO	<code>OIDC_GOOGLE_*</code> , <code>OIDC_MICROSOFT_*</code>
Resend / SMTP	Email	<code>EMAIL_*</code> / <code>IntegrationSettings</code>
FFmpeg	Transcode rekaman	binary di PATH
Storage local	Unggahan + rekaman meeting	<code>uploads/</code> (meeting belum S3)

4. Flowchart — Alur sistem LanPro

4.1 Peta modul end-to-end

```

flowchart LR
    subgraph Auth
        REG[Daftar] --> PEND[Pending]
        PEND --> APPR[Approve Admin]
        APPR --> LOGIN[Login / OIDC]
    end
    LOGIN --> HOME{Punya proyek?}
    HOME -->|Tidak| WEL[WelcomeScreen]
    HOME -->|Ya| DASH[Dashboard]
    DASH --> ISS[Daftar Isu]
    DASH --> PLAN[Planning / Roadmap]
    DASH --> KAN[Kanban]
    DASH --> QA[QA]
    DASH --> WIKI[Wiki]
    DASH --> FLOW[Flowchart]
    DASH --> MEET[Meeting]
    DASH --> TEAM[Tim]
    ISS --> DETAIL[Detail Isu + History]
    QA --> BUG[Bug Task linkedTaskId]
    MEET --> AI[Pipeline AI]
  
```

4.2 Siklus hidup akun

```

stateDiagram-v2
    [*] --> Registered: POST register
    Registered --> Pending: status pending
    Pending --> Approved: Admin approve + email
    Pending --> Rejected: Admin reject
    Approved --> ActiveSession: Login JWT
    ActiveSession --> ForcePasswordChange: Temp password / policy
    ForcePasswordChange --> ActiveSession: Ganti sandi
    ActiveSession --> LoggedOut: Logout / blacklist
  
```

4.3 Siklus hidup isu

flowchart TD

A[Buat isu / quick create] --> B[Backlog atau Sprint/Milestone]

B --> C[Kanban: status MasterData code]

C --> D{Done / Closed?}

D -->|Ya| E[History + notifikasi]

D -->|Tidak| C

F[QA FAIL] --> G[Bug Task]

G --> H[linkedTaskId + linkedBugKey]

H --> I[Bug Done → QA Retest]

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5. Flowchart — Agile vs Waterfall

5.1 Pemilihan metodologi

flowchart TD

```

A[Buat / Edit Proyek] --> B[Pilih methodology]
B --> C{normalisasi UPPERCASE}
C -->|AGILE / SCRUM / ...| D[Mode Agile]
C -->|WATERFALL| E[Mode Waterfall]
D --> D1[Tampil menu Planning & Sprint]
D --> D2[Label Roadmap & Timeline]
D --> D3[Burndown + satu sprint aktif + scope lock]
E --> E1[Sembunyikan menu Sprint]
E --> E2[Label Roadmap & Milestones]
E --> E3[API create sprint ditolak]
E --> E4[Redirect view sprints → timeline]

```

5.2 Alur Agile

flowchart TD

```

A[Backlog berisi isu] --> B[Buat Sprint planned]
B --> C[DnD isu ke sprint planned]
C --> D[Start Sprint → status active]
D --> E[Demote sprint aktif lain → planned]
D --> F[Lingkup terkunci]
F --> G{User DnD masuk/keluar?}
G -->|Tanpa unlockScope| H[Tolak 400]
G -->|Complete + unlockScope| I[Pindah sisa: backlog / next / leave]
I --> J[Sprint completed terkunci]
D --> K[Dashboard burndown hybrid]

```

5.3 Alur Waterfall

flowchart TD

A[Proyek WATERFALL] --> B[Tidak ada Planning Sprint di nav]

B --> C[Roadmap & Milestones]

C --> D[CRUD Milestone]

D --> E[Taut isu via milestoneId]

E --> F[Gantt + garis blocks]

A --> G[POST /sprints]

G --> H[Error metodologi_waterfall_tidak_mendukung]

5.4 Perbandingan cepat

Aspek	Agile LanPro	Waterfall LanPro	Jira (kasar)
Unit perencanaan	Sprint	Milestone + Gantt	Sprint / Version
Board	Kanban status dinamis	Kanban sama	Board
Scope lock	Ya (#461)	N/A	Sprint incomplete
Satu aktif	Ya (#462)	N/A	Umum praktik
CPM / baseline	Tidak	Tidak	Add-on

6. Flowchart — Per peran sistem

6.1 Administrator (`admin`)

flowchart TD

```
L[Login] --> A[God Mode semua proyek]
A --> B[User Management CRUD]
A --> C[Master Data CRUD]
A --> D[Audit / DB Explorer / Settings]
A --> E[Buat proyek]
E --> F[Assign anggota]
A --> G[Masuk modul proyek apa pun]
```

6.2 Department Head (`head` sistem)

flowchart TD

```
L[Login] --> A[Lihat user/master/audit/settings: R]
A --> B[Daftar proyek se-departemen]
B --> C[Hak di dalam proyek = peran ProjectMembers]
```

6.3 Standard User & Observer sistem

flowchart TD

```
L[Login] --> A{Anggota proyek?}
A -->|Tidak| W[WelcomeScreen]
A -->|Ya| P[Hak mengikuti peran proyek]
```

7. Flowchart — Per peran proyek

7.1 Owner / Project Admin / Manager

flowchart TD

```
O[Owner] --> O1[CRUD hampir semua + hapus proyek + setelan]
A[Project Admin] --> A1[CRUD modul + setelan + Tim CRUD]
A --> A2[Bukan God Mode sistem]
M[Manager] --> M1[CRUD operasional + Tim R+U]
M --> M2[Start/Complete sprint Agile]
```

7.2 System Analyst / Business Analyst

flowchart TD

```
SA[System Analyst] --> W[Wiki CRUD + Flowchart CRUD]
SA --> S2[List CRU · Board RU · Meeting CRU]
BA[Business Analyst] --> M[Meeting Notes CRUD + AI]
BA --> B2[Wiki/Flowchart CRU · List CRU]
```

7.3 Developer / QA / Head / Viewer

flowchart TD

```
DEV[Developer] --> D1[List CRU · Board RU · QA RU]
QA[QA] --> Q1[QA CRUD · List CRU · Meeting CRU]
QA --> Q2[Fail → Bug Task linkedTaskId]
H[head/viewer proyek] --> R[R semua modul]
```

8. Matriks akses

Sumber: MATRIKS_PROYEK / MATRIKS_SISTEM . Huruf = C R U D.

8.1 SISTEM

Modul	admin	head	user	viewer
userManagement	CRUD	R	—	—
masterData	CRUD	R	—	—
auditLog	CRUD	R	—	—
dbExplorer	CRUD	—	—	—
settings	CRUD	R	—	—
buat proyek	C	—	—	—

8.2 PROYEK (CUPLIKAN)

Modul	owner/admin/manager	SA	BA	developer	qa	head/viewer
dashboard	R	R	R	R	R	R
list	CRUD*	CRU	CRU	CRU	CRU	R
board	CRUD*	RU	RU	RU	RU	R
sprints / timeline	CRUD*	R	R	R	R	R
wiki	CRUD*	CRUD	CRU	R	R	R
flowchart	CRUD*	CRUD	CRU	R	R	R
meetingNotes	CRUD*	CRU	CRUD	R	CRU	R
qa	CRUD*	RU	RU	RU	CRUD	R
access	CRUD / RU†	R	R	R	R	R

* manager = CRUD operasional sesuai matriks penuh.

† access: owner/admin CRUD; manager R+U.

Setelan proyek: owner + project admin. Hapus proyek: owner saja.

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9. Spesifikasi per modul

Setiap subbagian mengikuti pola: **Bisnis** → **Fungsional** → **Teknis** → **Swimlane** → **Aturan**.

9.1 Auth & akun

Bisnis. Hanya akun **approved/active** yang boleh masuk. Admin menyetujui pendaftar. SSO Google/Microsoft setara password untuk domain yang dikonfigurasi.

Fungsional. Register, approve/reject, login, logout, forgot/reset password (kata sandi sementara 2 jam — #448/#451), wajib ganti sandi, lengkapi pendaftaran OIDC.

Teknis. `src/features/auth/` · `server/routes/auth.routes.ts` · `auth-oidc.routes.ts` · approve via `user.routes.ts` PUT status. JWT di klien; blacklist saat logout/forgot.

```
sequenceDiagram
    participant U as User
    participant A as Auth API
    participant Adm as Admin Users
    U->>A: POST /register
    A-->>U: pending
    Adm->>A: PUT /users/:id approved
    U->>A: POST /login ATAU OIDC start->callback
    A-->>U: JWT
```

Detail penuh: §11.

9.2 Proyek (buat, setelan, metodologi)

Bisnis. Hanya Administrator sistem yang membuat proyek. Owner/Admin proyek mengubah setelan & metodologi. Hapus proyek = hard cascade, owner saja.

Fungsional. Create project, set `category` /methodology (code MasterData, UPPERCASE), invite anggota, dashboard layout, delete.

Teknis. `server/routes/project.routes.ts` · `project-modules.routes.ts` · UI di `AppContainer` + chrome proyek. `bolehBuatProyek` / `bolehUbahSetelanProyek` / `bolehHapusProyek` .

flowchart TD

```

Admin[System admin] --> Create[POST /projects]
Create --> Meth[Set AGILE atau WATERFALL]
Meth --> Members[Invite + ProjectMembers.role]
Members --> Work[Masuk modul]
Owner --> Del[DELETE hard cascade]

```

9.3 Dashboard

Bisnis. Ringkasan kesehatan proyek: KPI isu, sprint aktif (Agile), aktivitas.

Fungsional. Widget KPI, filter sprint, burndown hybrid (#464), layout tersimpan.

Teknis. `src/features/dashboard/` · data agregat dari `tasks/sprints/meetings` · `PUT`
`.../dashboard-layout` · guard `dashboard` R.

flowchart LR

```

Open[Buka proyek] --> Dash[Dashboard]
Dash --> Agg[Agregasi tasks/sprints/meetings]
Agg --> KPI[KPI]
Agg --> Burn{Agile + points?}
Burn -->|Ya| Pts[Burndown poin]
Burn -->|Tidak| Cnt[Burndown jumlah tugas]

```

9.4 Daftar Isu / Task

Bisnis. Sumber kebenaran pekerjaan. Status/prioritas/tipe dari MasterData **code**. History field-diff ala Jira (#452/#456). Hapus permanen (#394) + cascade (#444).

Fungsional. List, filter, quick create, detail modal, komentar, lampiran, subtask/link, activity history, bulk delete.

Teknis. `src/features/issues/` · `server/routes/task.routes.ts` · upload `file.routes.ts` · Soft FK sprint/milestone; scope lock berlaku saat pindah sprint aktif/selesai.

flowchart TD

```
List[Issue list / quick create] --> Detail[Task detail]
Detail --> Fields[Update field + code MasterData]
Detail --> Att[Attachments]
Detail --> Hist[Activity History]
Detail --> Del[Hard delete + Swal]
Fields --> Board[Status juga di Kanban]
```

Swimlane peran (ringkas).

flowchart LR

```
subgraph Manage[owner/admin/manager]
    M1[CRUD penuh]
end
subgraph Build[SA/BA/dev/qa]
    B1[CRU list]
end
subgraph Read[head/viewer]
    R1[R saja]
end
```

9.5 Planning & Sprint (Agile)

Bisnis. Rencanakan iterasi; setelah start, lingkup terkunci agar burndown bermakna; hanya satu sprint aktif.

Fungsional. Backlog, buat sprint planned, DnD, Start, Complete (backlog/next/leave), badge terkunci, hapus sprint → tugas ke backlog dengan unlock.

Teknis. `src/features/planning/` · `sprints.routes.ts` · `sprintLingkup.ts` · Waterfall: UI disembunyikan + POST ditolak.

flowchart TD

```
Backlog --> Plan[Sprint planned]
Plan --> DnD[DnD tugas]
DnD --> Start[active + demote lain]
Start --> Lock[Scope lock]
Lock --> Complete[unlockScope + move sisa]
Complete --> Done[completed terkunci]
```

9.6 Papan Kanban

Bisnis. Visualisasi alur status kerja harian; WIP lunak sebagai sinyal overload (#455); blocker mencegah Done.

Fungsional. Kolom dari MasterData status; DnD; WIP badge; Socket.IO refresh.

Teknis. `src/features/kanban/` · PUT task status · normalisasi code case-insensitive (#459).

```
flowchart LR
    Card[Kartu] --> Drag[DnD ke kolom status]
    Drag --> Check{Blockers selesai?}
    Check -->|Tidak + Done| Reject[Toast tolak]
    Check -->|Ya| Save[PUT status code]
```

9.7 Timeline / Roadmap / Milestone

Bisnis. Perencanaan berbasis tanggal & milestone — permukaan utama Waterfall; tetap ada di Agile sebagai Roadmap & Timeline.

Fungsional. CRUD milestone, taut tugas (`milestoneId`), Gantt, garis dependensi `blocks` (#457). Bukan CPM.

Teknis. `src/features/timeline/` · `milestones.routes.ts` · label sidebar via `alurMetodologi` (#465).

```
flowchart TD
    WF[Proyek] --> TL[Timeline]
    TL --> MS[CRUD milestones]
    MS --> Link[Assign milestoneId]
    Link --> Gantt[Gantt + edges blocks]
```

9.8 QA

Bisnis. Suite uji → eksekusi → gagal menjadi bug yang dilacak sebagai Task agar Dev/QA satu rantai.

Fungsional. Suite/case CRUD, eksekusi Pass/Fail/..., history, generate AI (opsional), create bug Task, retest.

Teknis. `src/features/qa/` · `qa.routes.ts` · `linkedTaskId` + `linkedBugKey` (#463).

```

flowchart TD
    Suite --> Case
    Case --> Exec[Eksekusi]
    Exec -->|FAIL| Bug[Create Task bug]
    Bug --> FK[linkedTaskId + linkedBugKey]
    FK --> Dev[Dev di Board/List]
    Dev --> Retest[QA retest]

```

RBAC: peran `qa` = CRUD modul `qa`.

9.9 Wiki (Dokumentasi)

Bisnis. Basis pengetahuan proyek; System Analyst pemilik domain CRUD.

Fungsional. CRUD dokumen (teks/file/link), download, hard delete, sanitasi teks (#348).

Teknis. `src/features/wiki/` · `documents.routes.ts` dengan guard modul `wiki`.

```

flowchart LR
    List[Wiki list] --> Create[POST document]
    Create --> Edit[PUT content/file]
    Edit --> DL[Download]
    Edit --> Del[DELETE hard]

```

9.10 Flowchart

Bisnis. Diagram alur requirement/proses; System Analyst pemilik CRUD.

Fungsional. Buat diagram, edit nodes/edges, import, simpan `canvasData` JSON (#136), hapus.

Teknis. `src/features/flowchart/` · penyimpanan lewat Documents API (`type: flowchart`).

Catatan teknis (utang): UI matriks memakai modul `flowchart` ; guard server dokumen saat ini berkunci ke `wiki` . Perilaku efektif: SA tetap lolos CRUD dokumen; pemisahan guard flowchart vs wiki belum sempurna di API.

flowchart TD

```
Dash[Flowchart dashboard] --> New[POST type=flowchart]
New --> Canvas[Edit nodes/edges]
Canvas --> Save[PUT canvasData]
Save --> Del[DELETE]
```

9.11 Tim / Access

Bisnis. Siapa anggota proyek dan peran apa — mengunci seluruh matriks modul.

Fungsional. Invite, ubah role (code MasterData `project_role`), hapus anggota.

Teknis. `src/features/team/` · `members/invites` di `project.routes.ts` · modul matriks `access`.

flowchart TD

```
Team[Panel Tim] --> Invite
Invite --> Role[Set ProjectMembers.role code]
Role --> Matrix[jagaProyek enforce matriks]
```

9.12 Administrasi sistem

Submodul	Bisnis	UI	API
Master Data	Katalog status/peran/dept/methodology	<code>features/master</code>	<code>master-data.routes.ts</code>
Users	Approve, CRUD user, sesi	<code>features/users</code>	<code>user.routes.ts</code> , <code>session.routes.ts</code>
Audit	Jejak immutable	<code>features/enterprise-audit</code>	<code>audit.routes.ts</code>
Settings	Email/WA/sistem	<code>features/settings</code>	<code>system.routes.ts</code>
DB Explorer	Baca skema/SQL admin	<code>features/explorer</code>	<code>db-admin.routes.ts</code>

flowchart TD

```

SysAdmin[Users.role=admin] --> MD[Master Data]
SysAdmin --> US[Users + sessions]
SysAdmin --> AU[Audit]
SysAdmin --> ST[Settings]
SysAdmin --> DB[DB Explorer]
Head[head sistem] --> R[Read-only admin surfaces]

```

9.13 Catatan Rapat (modul induk Meeting AI)

Bisnis. BA mengelola notulen; poin diskusi; AI mengubah rekaman/teks → ringkasan terstruktur. Detail pipeline: §10.

Fungsional. CRUD meeting, discussion points + komentar, AI companion (live/upload/paste), unduh lampiran.

Teknis. `src/features/meeting-notes/` · `meetings.routes.ts` · `discussion-points.routes.ts`.
List **tidak** boleh import berat FFmpeg (#460).

10. Meeting AI — spesifikasi dalam

10.1 BRD Meeting AI

Aspek	Isi
Nilai bisnis	Mengurangi waktu BA menulis notulen; mengekstrak keputusan/action item dari audio/video/teks
Pengguna utama	<code>business_analyst</code> (CRUD); owner/admin/manager CRUD; SA/qa CRU; dev/head/viewer R
Input	(1) rekam live mic (2) unggah audio/video ≤120MB (3) paste / <code>.txt</code> manuscript
Output	Transcript + <code>aiSummary</code> terstruktur; file rekaman dihapus setelah sukses
Bukan janji	Parity Otter (diarization speaker belum); SLA waktu real-time tetap
Status produk	Kode gelombang 1 + hardening ada; E2E COMPLETED pemilik = #320 BELUM

10.2 FSD — tiga jalur input

Jalur	Langkah pengguna	Hasil
Live	Buka detail → AiMeetingCompanion → rekam ≥5s ≥1KB → proses	Pipeline async
Upload	Pilih media → (opsional) FFmpeg.wasm → MP3 → chunk upload	Pipeline auto-start
Manuscript	Paste / <code>.txt</code> ≤5MB	<code>analyze-transcript</code> sync (tanpa STT file)

Poin diskusi: CRUD terpisah; authorId dari JWT (#251). Hapus meeting cascade points + file (#444).

Aturan ekstra: PUT/DELETE meeting hanya **author atau system ADMIN** (ketat vs matriks U/D).

10.3 TSD — komponen & API

Lapisan	Path
List shell	MeetingNotes.tsx
Detail + points	DiscussionPointsTable.tsx (lazy #460)
AI UI	AiMeetingCompanion.tsx
Live helpers	lib/liveRecording.ts
Pipeline server	server/services/meeting.service.ts (runAIPipeline)
Transcode	server/services/meeting-transcode.ts
Filter anti-halusinasi	server/services/meeting-ai-filter.ts
Gemini wrapper	server/services/ai.service.ts

API UTAMA

Method	Path	Guard	Catatan
POST	/api/v1/meetings/:id/upload-recording	meetingNotes U	Chunk merge; auto runAIPipeline
GET	/api/v1/meetings/:id/status	R	Poll progress
POST	/api/v1/meetings/:id/cancel	U	Hapus file → IDLE
POST	/api/v1/meetings/:id/analyze	U	Manual 202 (jarang dipakai UI)
POST	.../analyze-transcript	U	Paste path
POST	.../analyze-video	U	Multimodal 2.5 Pro (UI belum panggil)
CRUD	/api/projects/:pid/meetings	C/R/U/D	+ discussion points

ENV & DEPENDENSI

Var / dep	Fungsi
<code>GEMINI_API_KEY</code>	STT + LLM
<code>AI_PIPELINE_MAX_CONCURRENT</code>	Default 2 (clamp 1–4), in-process (#322)
<code>ffmpeg</code> di PATH	Transcode server
Disk <code>uploads/</code>	Rekaman lokal — bukan <code>STORAGE_DRIVER</code> S3

10.4 State machine pipeline

```
stateDiagram-v2
    [*] --> IDLE
    IDLE --> UPLOAD_SUCCESS: upload selesai
    UPLOAD_SUCCESS --> EXTRACTING_AUDIO: ~15%
    EXTRACTING_AUDIO --> TRANSCRIBING_STT: ~60%
    TRANSCRIBING_STT --> ANALYZING_LLM: ~90%
    ANALYZING_LLM --> COMPLETED: hapus file disk
    ANALYZING_LLM --> FAILED: file dipertahankan
    IDLE --> IDLE: cancel
    EXTRACTING_AUDIO --> IDLE: cancel
    TRANSCRIBING_STT --> IDLE: cancel
    ANALYZING_LLM --> IDLE: cancel
```

Alias legacy: `PROCESSING_AI` → dipetakan ke `EXTRACTING_AUDIO` di GET status;
`TRANSCRIBING` → `TRANSCRIBING_STT`.

10.5 Swimlane end-to-end (live/upload)

```
sequenceDiagram
    participant BA as BA / User
    participant UI as AiMeetingCompanion
    participant FE as FFmpeg.wasm
    participant API as meetings.routes
    participant Pipe as runAIPipeline
    participant FF as FFmpeg CLI
    participant G as Gemini
    participant DB as PostgreSQL
    participant Disk as uploads/

    BA->>UI: Rekam / pilih file
    UI->>FE: Transcode MP3 (best effort)
    FE-->>UI: MP3 atau raw fallback #447
    UI->>API: POST upload-recording chunks
    API->>Disk: merge file
    API->>DB: UPLOAD_SUCCESS + recording_url
    API->>Pipe: fire-and-forget
    Pipe->>FF: extract audio MP3
    Pipe->>DB: EXTRACTING_AUDIO
    Pipe->>G: STT
    Pipe->>DB: TRANSCRIBING_STT
    Pipe->>G: structured LLM
    Pipe->>DB: ANALYZING_LLM → COMPLETED + aiSummary
    Pipe->>Disk: delete recording
    Pipe-->>UI: Socket meeting_ai_completed / poll status
    BA->>UI: Baca ringkasan + transcript
```

10.6 Swimlane manuscript (paste)

```
sequenceDiagram
    participant BA as BA
    participant UI as UI
    participant API as analyze-transcript
    participant G as Gemini flash
    participant DB as DB

    BA->>UI: Paste / .txt
    UI->>API: POST analyze-transcript
    API->>G: analisis + filter UNVERIFIED
    API->>DB: transcript + aiSummary
    API-->>UI: hasil sync
```

10.7 Progress & kegagalan

- Socket: `meeting_ai_status`, `meeting_ai_completed`, `meeting_ai_failed`
- Fallback poll 3s pada `/status` setelah upload yang memulai pipeline
- Cancel: hapus disk dulu, lalu `IDLE` (#441) — pipeline berhenti antar tahap
- FAILED: file tetap untuk retry
- COMPLETED: file dihapus (retensi #320)

10.8 Gap & risiko (#320)

Gap	Dampak
E2E pemilik belum COMPLETED	Jangan klaim production-ready Meeting AI
Tanpa diarization	Speaker tidak terpisah
Disk lokal / Vercel ephemeral	Benturan #30
Semaphore in-process saja	Bukan antrian durable multi-instance
<code>analyze-video</code> belum di UI	Kapabilitas server idle

11. Auth & OIDC — spesifikasi dalam

11.1 BRD Auth

Aspek	Isi
Tujuan	Identitas terpercaya sebelum akses proyek
Saluran	Email+password · Google OIDC · Microsoft OIDC (<code>bni.co.id</code> target #305)
Onboarding	Self-register → pending → admin approve → email aktif (#261 terkait)
Pemulihan	Forgot → kata sandi sementara terbatas waktu → wajib ganti

11.2 FSD Auth

1. Register → `pending`
2. Admin set `approved` / `active` / `rejected`
3. Login password ATAU SSO
4. SSO user baru → lengkapi pendaftaran bila perlu
5. Forgot password → email → login temp → force change
6. Logout → blacklist JWT

11.3 TSD Auth

Area	Path
UI	<code>src/features/auth/</code> (<code>LoginScreen</code> , <code>RegisterScreen</code> , <code>SsoButtons</code> , <code>forgot/reset modals</code>)
Password API	<code>server/routes/auth.routes.ts</code>
OIDC API	<code>server/routes/auth-oidc.routes.ts</code>
Adaptor	<code>server/services/oidc.service.ts</code> (discovery Microsoft/Google)
Approve	<code>server/routes/user.routes.ts</code>

OIDC FLOW

```
sequenceDiagram
    participant U as User
    participant FE as SsoButtons
    participant API as auth-oidc
    participant IdP as Google/Microsoft
    U->>FE: Klik provider
    FE->>API: GET /:provider/start
    API-->>U: Redirect IdP
    U->>IdP: Login consent
    IdP->>API: callback + code
    API->>IdP: token + claims
    API->>API: upsert/link user
    API-->>FE: sso_token / session
    Note over API,FE: Microsoft: OIDC_MICROSOFT_* + tenant (#305 verifikasi lapangan)
```

PASSWORD FORGOT (RINGKAS TERKINI)

```
flowchart TD
    A[Forgot password] --> B[Email terdaftar?]
    B -->|Tidak| C[404 email_tidak_terdaftar]
    B -->|Ya| D[Set passwordHash sementara + blacklist sesi]
    D --> E[Email kata sandi temp 2 jam]
    E --> F[Login → wajib ganti sandi]
```

11.4 Gap Auth

Gap	Tiket
Verifikasi login Microsoft sungguhan domain BNI	#305
Arah httpOnly cookie vs localStorage JWT	terkait #30 keamanan

12. Lampiran

12.1 Artefak terkait

Artefak	Path
Skema DB	docs/DATABASE_SCHEMA.md
API	docs/API-SURFACE.md , docs/openapi.json
Keamanan	docs/security_spec.md
Papan	AUDIT.md §1, §19, §24
Aturan agen	AGENTS.md
PDF (setelah generate)	docs/LanPro-BRD-FSD-TSD-v3.1.pdf

12.2 Yang dokumen ini bukan

- Bukan kontrak hukum / SLA pelanggan.
- Bukan OpenAPI penuh setiap rute.
- Bukan janji fitur out-of-scope (#346) atau parity Otter.

12.3 Riwayat versi

Ver	Tanggal	Perubahan
2.5	sebelumnya	Draft enterprise; RBAC satu tingkat — basi
3.0	06 Sep 2026	Two-Tier RBAC, dual-mode, flowchart per role
3.1	06 Sep 2026	Per modul BRD+FSD+TSD+swimlane; Meeting AI dalam; Auth/OIDC dalam; #466

12.4 Keputusan review tersisa

1. Apakah utang guard Flowchart= `wiki` di API perlu tiket baru (mis. #467)?
2. Setelah review teks: cabut watermark DRAFT pada PDF berikutnya?
3. Tutup formal #4

Akhir dokumen v3.1.0 — siap review pemilik · tiket #466.

DRAFT